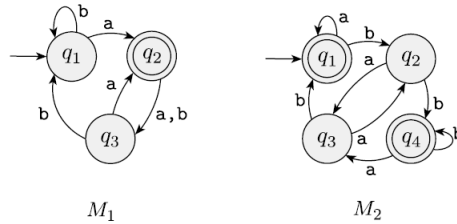


CSE 309 Theory of Automata, Spring 2026

Exercise 2

- 1.1 The following are the state diagrams of two DFAs, M_1 and M_2 . Answer the following questions about each of these machines.



- a. What is the start state?
 - b. What is the set of accept states?
 - c. What sequence of states does the machine go through on input aabb?
 - d. Does the machine accept the string aabb?
 - e. Does the machine accept the string ε ?
- 1.2 Give the formal description of the machines M_1 and M_2 pictured in Exercise 1.1.
- 1.3 The formal description of a DFA M is $(\{q_1, q_2, q_3, q_4, q_5\}, \{u, d\}, \delta, q_3, \{q_3\})$, where δ is given by the following table. Give the state diagram of this machine.

	u	d
q_1	q_1	q_2
q_2	q_1	q_3
q_3	q_2	q_4
q_4	q_3	q_5
q_5	q_4	q_5

- 1.4 Each of the following languages is the intersection of two simpler languages. In each part, construct DFAs for the simpler languages, then combine them using the construction discussed in footnote 3 (page 46) to give the state diagram of a DFA for the language given. In all parts, $\Sigma = \{a, b\}$.
- a. $\{w \mid w \text{ has at least three a's and at least two b's}\}$
 - ^Ab. $\{w \mid w \text{ has exactly two a's and at least two b's}\}$
 - c. $\{w \mid w \text{ has an even number of a's and one or two b's}\}$
 - ^Ad. $\{w \mid w \text{ has an even number of a's and each a is followed by at least one b}\}$
 - e. $\{w \mid w \text{ starts with an a and has at most one b}\}$
 - f. $\{w \mid w \text{ has an odd number of a's and ends with a b}\}$
 - g. $\{w \mid w \text{ has even length and an odd number of a's}\}$

1.5 Each of the following languages is the complement of a simpler language. In each part, construct a DFA for the simpler language, then use it to give the state diagram of a DFA for the language given. In all parts, $\Sigma = \{a, b\}$.

- ^Aa. $\{w \mid w \text{ does not contain the substring } ab\}$
- ^Ab. $\{w \mid w \text{ does not contain the substring } baba\}$
- c. $\{w \mid w \text{ contains neither the substrings } ab \text{ nor } ba\}$
- d. $\{w \mid w \text{ is any string not in } a^*b^*\}$
- e. $\{w \mid w \text{ is any string not in } (ab^+)^*\}$
- f. $\{w \mid w \text{ is any string not in } a^* \cup b^*\}$
- g. $\{w \mid w \text{ is any string that doesn't contain exactly two } a\text{'s}\}$
- h. $\{w \mid w \text{ is any string except } a \text{ and } b\}$

1.6 Give state diagrams of DFAs recognizing the following languages. In all parts, the alphabet is $\{0,1\}$.

- a. $\{w \mid w \text{ begins with a } 1 \text{ and ends with a } 0\}$
- b. $\{w \mid w \text{ contains at least three } 1\text{'s}\}$
- c. $\{w \mid w \text{ contains the substring } 0101 \text{ (i.e., } w = x0101y \text{ for some } x \text{ and } y)\}$
- d. $\{w \mid w \text{ has length at least } 3 \text{ and its third symbol is a } 0\}$
- e. $\{w \mid w \text{ starts with } 0 \text{ and has odd length, or starts with } 1 \text{ and has even length}\}$
- f. $\{w \mid w \text{ doesn't contain the substring } 110\}$
- g. $\{w \mid \text{the length of } w \text{ is at most } 5\}$
- h. $\{w \mid w \text{ is any string except } 11 \text{ and } 111\}$
- i. $\{w \mid \text{every odd position of } w \text{ is a } 1\}$
- j. $\{w \mid w \text{ contains at least two } 0\text{'s and at most one } 1\}$
- k. $\{\epsilon, 0\}$
- l. $\{w \mid w \text{ contains an even number of } 0\text{'s, or contains exactly two } 1\text{'s}\}$
- m. The empty set
- n. All strings except the empty string

1.7 Give state diagrams of NFAs with the specified number of states recognizing each of the following languages. In all parts, the alphabet is $\{0,1\}$.

- ^Aa. The language $\{w \mid w \text{ ends with } 00\}$ with three states
- b. The language of Exercise 1.6c with five states
- c. The language of Exercise 1.6l with six states
- d. The language $\{0\}$ with two states
- e. The language $0^*1^*0^+$ with three states
- ^Af. The language $1^*(001^+)^*$ with three states
- g. The language $\{\epsilon\}$ with one state
- h. The language 0^* with one state